VELOOP REWARDS

Product Analytics, Data Readiness & Data Management Blueprint

Data Analyst Internship, Task 01. Final Consolidated Report, version 1.0, 31 October 2026

DATA AUTHENTICITY NOTICE. VELOOP is a real product under development, but all data analysed in this report is fictional/simulated (SIMULATED DATA - CREATED FOR ANALYTICAL TESTING). No figure here describes actual VELOOP users, earnings, withdrawals, revenue, retention or performance. Every event, field and KPI is Proposed or To Be Verified unless marked Confirmed.

Contents of this deliverable pack

Required deliverable	Where
01 Executive summary	Section 1
02 Product data audit	Section 3; workbook sheets 01-03
03 Data coverage matrix	Workbook sheet 02
04 User analytics specification	Section 4; sheet 04
05-07 Earning/reward, wallet/withdrawal, referral/engagement	Sections 5-7
08 Event tracking plan	Section 8; sheet 05
09 Data dictionary	Sheet 06 (100 fields)
10 Logical data model	Section 9; figure er_diagram.png
11 KPI catalogue	Section 10; sheet 08 (46 KPIs)
12 Simulated data profiling	Section 11; Data_Profiling_and_DQ_Audit.xlsx
13 Data quality audit	Section 12; sheet K_DQ_Report (34 issues)
14 Dashboard blueprint	Section 13; wireframes (figures/)
15 Developer handoff	Section 14; sheet 12
16 Pipeline proposal	Section 15
17 Privacy & governance	Section 16
18 Readiness assessment	Section 17
19 Assumption register	Section 18; sheet 16
20 Implementation backlog	Section 19; sheet 17
21 Final consolidated report	This document
Code, tests, cleaning	src/, tests/, README.md

1. Executive summary

What was investigated

The brief's product scope (accounts, earning tasks, rewards, wallet, withdrawals, referrals, engagement and support) was turned into a complete analytics foundation: a data-readiness audit, a 38-event tracking plan, a 100-field data dictionary, a logical data model, 46 KPIs, quality controls, dashboard specifications, an engineering handoff, a pipeline, governance rules, a change process and a prioritised backlog. Two simulated workbooks were then profiled and audited with reusable, tested code.

What was discovered

- **Two Users tables contradict each other.** Both files use the same 1,000 user IDs (VLUSR000001 to VLUSR001000), yet names, emails and registration times differ for all 1,000 users. One table must be declared the user master before any join.
- **The supplied pack is incomplete.** Rewards, Wallet, Withdrawals and Referrals tables, which the brief lists, were not supplied. The behaviour file contains users, sessions, 35,704 events, daily activity, task activity and milestones only.
- **Structure is sound but not production-ready.** No orphan user or session references exist, but task attempts carry no reward amount or unit, the task catalogue is missing (every task_id maps to several categories), and stored counts do not reconcile (events_in_session wrong for 7,537 of 7,913 sessions; 423 daily rows with more completed than total tasks).
- **The messy sheet behaves as designed.** Automated checks found 8 duplicate user rows, 9 missing required values, 1 malformed date and 2 conflicting registration times; the event audit sheet holds 10 duplicate events, 4 trailing-space statuses and 5 inconsistent feature names.
- **Personal data is over-collected.** Full name, email, age, gender and city are present for every user, contrary to the brief's data-minimisation rule.

What is currently unknown

Whether any event, session ID, transaction ID, reward pending state, withdrawal state history, acquisition source or the Spin/Mining/Missions/Badges features exist in production. 18 such unknowns are in the assumption register; none has been converted into a fact.

What VELOOP should implement first

- A stable user_id and one user master table.
- An immutable transaction_id with idempotent wallet writes, and separate reward-versus-activity records.
- Defined value units (VE, XP, Gems, Tokens) with conversion rules, so amounts are never summed across units.
- Withdrawal state and timestamp history; UTC timestamps everywhere.
- The Critical events REQ-001 to REQ-013, validated at ingestion by the proposed rules.

Major risks

Reporting activity as earnings; summing across units; duplicate credits; unreliable daily metrics without a UTC standard; third-party outcomes outside VELOOP's control; and silently mixing simulated with production data.

Major priorities

Six Critical, seven High, five Medium and two Low backlog items (Section 19). The Critical items are all Prelaunch.

Key figures from this work (simulated data and design artefacts)

Measure	Value
Proposed events / data-dictionary fields / KPIs	38 / 100 / 46
Dashboards / visuals / developer requirements	7 / 28 / 15

Measure	Value
Quality rules designed / automated tests passing	20 / 11 of 11
Simulated datasets profiled (sheets) / records in largest	9 / 35,714 (event audit sheet)
Data quality issues found (Critical / High / Medium / Low)	3 / 11 / 15 / 5 (34 in total; see Section 12)
Referential-integrity orphans between users, sessions, events, tasks	0
Proposed KPI groups testable on supplied data (yes / partial / no)	10 / 3 / 5
Open unknowns in the assumption register	18, none converted into a fact

Recommended roadmap

- **Before launch (Critical and High):** user master and stable user_id; transaction_id and idempotent wallet writes; unit and conversion rules; reward-versus-activity separation; withdrawal state history; UTC timestamps; ingestion validation; role-based access to curated tables.
- **At launch:** emit events REQ-001 to REQ-013; run the quality checks and the reward-to-wallet reconciliation (K29) daily; certify the analytical tables that power Dashboards 1, 5 and 6.
- **After launch:** session identifier, acquisition source capture, support categories, engagement events for Spin, Missions and Streaks; remaining dashboards; replace simulated data with governed production data, never mixing them.

Decisions needed from the team

- Which of the two Users tables is the user master (assumption A016).
- Whether the behaviour workbook ('Task 02') is intended input for this task.
- Whether Rewards, Wallet, Withdrawals and Referrals datasets exist and can be supplied.
- The qualifying-activity list that defines DAU and the referral activation rule (A017, A010).

2. Scope, method and evidence discipline

Method: (1) read the brief and record each product fact in a Source & Verification Register with status Confirmed, Proposed or To Be Verified; (2) design the analytics model from the product areas; (3) profile and audit the two supplied workbooks with a Python package whose checks are small functions and are covered by automated tests; (4) test whether each proposed KPI can actually be calculated from what was supplied; (5) consolidate.

ID	Information	Source Type	Status
S001	VELOOP is described as a developing online rewards/engagement ecosystem	Internal	Confirmed
S002	Product areas listed: accounts, earning activities, VEs, XP, Gems, Tokens, Spins, daily rewards, streaks, tasks, missions, badges, levels, referrals, wallet, exchange, withdrawals, support, future features	Internal	To Be Verified
S003	Supplied dataset is fictional/simulated (Users, Task Activity, Rewards, Wallet, Withdrawals, Referrals, Feature Events, Messy Data)	Simulated	Confirmed
S004	Supplied files: VELoop_Rewards_Data_01.xlsx (Users) and VELOOP_Simulated_User_Behavior_1000.xlsx (behaviour, labelled Task 02)	Simulated	Confirmed
S005	Whether any analytics events, IDs, or tables exist in production	Internal	To Be Verified
S006	Currency/units: VEs, XP, Gems, Tokens exist as value units	Internal	To Be Verified
S007	Withdrawal lifecycle Requested > Processing > Completed (+Rejected/Reversed)	Internal	Proposed
S008	Event names, naming convention, field names in this workbook	Internal	Proposed
S009	Public web search for official VELOOP Rewards product documentation	Official (search)	To Be Verified

Naming standard. Events use object_action in lowercase snake_case and past tense (task_completed, withdrawal_rejected). One object, one action, no feature prefixes: the feature is a property. This matches the brief's examples, sorts by object, and is easy for engineers to implement and test. Every event carries a common envelope: event_id, event_name, event_timestamp_utc, user_id, schema_version, platform, app_version, session_id (if available), source_system.

3. Product data audit and coverage

31 product-area rows are mapped (user action, system action, business outcome, data required, proposed event, source, business question, status, priority). Events are classed as User Event (the user did something), System Event (the system recorded an outcome) or Business Outcome (a derived result computed in the analytics layer, for example a converted referral).

Product Area	Feature	Proposed Event	Status	Priority
Registration	Account creation	user_registered	To Be Verified	Critical
Login	Authentication	user_logged_in	To Be Verified	Critical
Login	Logout	user_logged_out	Proposed	Low
Onboarding	Onboarding flow	onboarding_started / onboarding_step_completed / onboarding_completed	Proposed	High
Account lifecycle	Status management	account_status_changed	Proposed	High
Tasks	Task catalogue	task_viewed	Proposed	High
Tasks	Task open	task_opened	Proposed	Medium
Tasks	Task start	task_started	Proposed	Critical
Tasks	Task completion	task_completed	Proposed	Critical
Tasks	Task failure	task_failed	Proposed	High
Ads / third-party earning	Watch Ads	task_completed (task_type=ad) + provider_ref	To Be Verified	High
Rewards	Reward credit	reward_credited	Proposed	Critical
Rewards	Reward pending	reward_pending	Proposed	High
Rewards	Reward reversal	reward_reversed	Proposed	Critical
Rewards	Reward correction	reward_corrected	Proposed	Medium
Daily Rewards	Daily claim	daily_reward_claimed	To Be Verified	High
Streaks	Streak tracking	streak_updated	To Be Verified	Medium
Missions	Mission progress	mission_started / mission_completed	To Be Verified	Medium
Badges / Levels / XP	Progression	badge_earned	To Be Verified	Low
Spin	Spin game	spin_started / spin_completed	To Be Verified	Medium
Tap & Earn / Mining	Passive/active earn	task_* or feature-specific events	To Be Verified	Medium
Referrals	Referral creation	referral_created	Proposed	High
Referrals	Referred sign-up	referral_registered	Proposed	High

Product Area	Feature	Proposed Event	Status	Priority
Referrals	Activation / conversion	referral_activated / referral_converted	Proposed	High
Wallet	Ledger entry	wallet_transaction_created / _completed / _reversed	Proposed	Critical
Exchange	Unit exchange	wallet_transaction_created (type=exchange)	To Be Verified	Medium
Withdrawals	Request	withdrawal_requested	Proposed	Critical
Withdrawals	Processing/Decision	withdrawal_processing / _completed / _rejected / _reversed	Proposed	Critical
Help Center	Help content	help_article_viewed / support_request_created	Proposed	Medium
Sessions	App session	session_started	Proposed	High
Future features	Unlisted	-	To Be Verified	Low

Fifteen analytics gaps

Gap ID	Gap	Priority	Reason
G01	No verified stable user_id across systems	Critical	Every KPI needs unique-user counting
G02	No verified wallet transaction_id/idempotency key	Critical	Ledger integrity and money-like reporting
G03	Reward credited vs activity occurred not verified as separate records	Critical	Core brief requirement (C3)
G04	No withdrawal state/timestamp history verified	Critical	Withdrawals are a major product-data area
G05	Task validation status/third-party callback data not verified	High	External systems may differ from VELOOP records
G06	Failure classification does not exist/unknown	High	Needed for C2; proposal only
G07	Referral activation rule undefined	High	Brief E: distinguish stages
G08	Session identifier existence unknown	High	Cannot assume it is stored
G09	Event timestamp standard (UTC, client vs server) unknown	High	Daily KPIs depend on day boundaries
G10	Acquisition source capture unknown	Medium	Acquisition dashboards need it
G11	Account status change history unknown	High	Lifecycle analysis
G12	Unit definitions (VE/XP/Gems/Tokens) and conversions unknown	Critical	Mixing units gives meaningless totals
G13	No data dictionary / schema versioning in place	Medium	Engineering and analyst alignment
G14	Support/help content tracking unknown	Medium	Brief section I
G15	Feature-level event coverage for Spin/Mining/Tap unknown	Medium	Features may not be implemented

4. User and account analytics specification

Identifiers: user_id (stable join key across everything; immutable, never reused), event_id (deduplication of events), session_id (groups activity; its existence is To Be Verified), transaction_id (ledger deduplication and reconciliation). No names, emails, phone numbers or addresses are required for analytics.

Field	Type	Required?	Purpose	Status
user_id	string (UUID/opaque)	Yes	Stable join key across all entities	To Be Verified
registered_at	timestamp UTC	Yes	Cohorting, lifecycle	Proposed
registration_source	string enum	Yes (default 'unknown')	Acquisition analysis	Proposed
referral_id_at_signup	string	No	Link to referral	Proposed
platform	string enum	Yes	Platform mix	Proposed
app_version	string	Optional	Release impact	Proposed
country_code	string (ISO-3166 alpha-2)	Optional	Coarse geography; no address needed	Proposed
account_status	string enum	Yes	Lifecycle state	To Be Verified
status_changed_at	timestamp UTC	Conditional	Lifecycle timing	Proposed
onboarding_started_at	timestamp UTC	Optional	Onboarding funnel	Proposed
onboarding_completed_at	timestamp UTC	Optional	Activation	Proposed
first_activity_at	timestamp UTC	Derived	Time-to-first-action	Proposed
last_activity_at	timestamp UTC	Derived	Inactivity	Proposed
last_login_at	timestamp UTC	Optional	Returning-user logic	Proposed
session_id	string	Optional	Session grouping	To Be Verified

5. Earning and reward analytics

Lifecycle measured: Discovery (task_viewed) > Start (task_started) > Completion (task_completed, emitted only after the applicable validation) > Validation (validation_status) > Reward (reward_pending, reward_credited). Activity occurring and a reward actually being credited are two separate records, linked by source_activity_id; the KPI 'activity-to-credit rate' (K24) exposes any gap between them.

- Failed activity is classified with a Proposed category list: user_abandoned, tracking_failed, validation_failed, provider_error, system_error, unknown. These are not existing VELOOP codes.
- Reward events: reward_credited, reward_pending, reward_reversed (references original_reward_id), reward_corrected (carries amount_delta).
- Third parties: ads, surveys, offers and games may fail, change or charge back outside VELOOP's control, so provider and provider_ref are recorded and provider-caused outcomes are reported separately.

6. Wallet and withdrawal analytics

Wallet transaction model: transaction_id, user_id, timestamp, transaction_type, amount (always positive; direction gives the sign), unit, status, source, reference_id, related_transaction_id (reversal/correction chain). Transaction types reward_credit, redemption, reversal, adjustment and exchange are all Proposed.

Withdrawal lifecycle: requested > processing > completed, with rejected and reversed as alternatives. Fields: withdrawal_id, user_id, requested_amount, final_amount, method (a category, never account details), requested_at, processing_started_at, finalized_at, status, reason_category. Withdrawal KPIs are K30 to K38 in the catalogue (requests, completed, pending queue, rejected, reversed, processing time, completion rate, volume, method distribution), each with definition, formula, data required, source, frequency, stakeholder, purpose and limitation.

7. Referral and engagement analytics

Referral stages are kept separate: created > registered > activated > converted. Registration never counts as conversion; the activation rule is an open question (A010). Engagement is measured through DAU/WAU/MAU, returning users, feature adoption, repeat usage, task activity and stickiness (K03 to K05, K08, K15 to K17), all defined over a documented list of qualifying events.

8. Event tracking plan

38 events are catalogued with trigger, definition, class, properties, data type, validation, deduplication, source, purpose, KPI dependency and implementation status ('Proposed - To Be Verified' for all). The table below lists them; the complete specification is sheet 05.

Event Name	Class	Feature	Required Properties	Priority
user_registered	System Event	account	registration_source, platform	Critical
user_logged_in	User Event	account	platform	Critical
user_logged_out	User Event	account	platform	Low
session_started	User Event	app	platform	High
onboarding_started	User Event	onboarding	platform	High
onboarding_step_completed	User Event	onboarding	step_name, step_number	Medium
onboarding_completed	User Event	onboarding	platform	High
account_status_changed	System Event	account	old_status, new_status, change_actor_type	High
task_viewed	User Event	tasks	task_id, task_type, feature	Medium
task_opened	User Event	tasks	task_id, task_type, feature	Medium
task_started	User Event	tasks	task_activity_id, task_id, task_type, feature	Critical
task_completed	System Event	tasks	task_activity_id, task_id, task_type, feature, status,	Critical

Event Name	Class	Feature	Required Properties	Priority
			validation_status, reward_amount, reward_unit	
task_failed	System Event	tasks	task_activity_id, task_id, failure_category	High
reward_pending	System Event	rewards	reward_id, user_id, source_activity_id, amount, unit, pending_reason	High
reward_credited	System Event	rewards	reward_id, source_activity_id, amount, unit, transaction_id, feature	Critical
reward_reversed	System Event	rewards	reward_id, original_reward_id, amount, unit, reason_category	Critical
reward_corrected	System Event	rewards	reward_id, original_reward_id, amount_delta, unit, reason_category	Medium
daily_reward_claimed	User Event	daily_rewards	claim_date, reward_id	High
streak_updated	System Event	streaks	streak_length, change_type(increment|reset)	Medium
mission_started	User Event	missions	mission_id	Medium
mission_completed	System Event	missions	mission_id	Medium
badge_earned	System Event	progression	badge_id	Low
spin_started	User Event	spin	spin_id	Medium
spin_completed	System Event	spin	spin_id, outcome_unit, outcome_amount	Medium
referral_created	User Event	referrals	referral_id, referrer_user_id	High
referral_registered	System Event	referrals	referral_id, referred_user_id	High
referral_activated	System Event	referrals	referral_id, referred_user_id, activation_rule_id	High
referral_converted	System Event	referrals	referral_id, conversion_rule_id	High
wallet_transaction_created	System Event	wallet	transaction_id, transaction_type, amount, unit, status, reference_id	Critical
wallet_transaction_completed	System Event	wallet	transaction_id, completed_at	Critical
wallet_transaction_reversed	System Event	wallet	transaction_id, related_transaction_id, reason_category	Critical
withdrawal_requested	User Event	withdrawals	withdrawal_id, requested_amount, unit, method	Critical
withdrawal_processing	System Event	withdrawals	withdrawal_id	Critical
withdrawal_completed	System Event	withdrawals	withdrawal_id, final_amount, method	Critical
withdrawal_rejected	System Event	withdrawals	withdrawal_id, reason_category	Critical
withdrawal_reversed	System Event	withdrawals	withdrawal_id, reason_category	High
support_request_created	User Event	support	ticket_id, issue_category	Medium
help_article_viewed	User Event	help	article_id, category	Medium

9. Logical data model

Ten entities, each with a primary key and foreign keys. Solid entities exist in the supplied simulated data; dashed entities are proposed only.

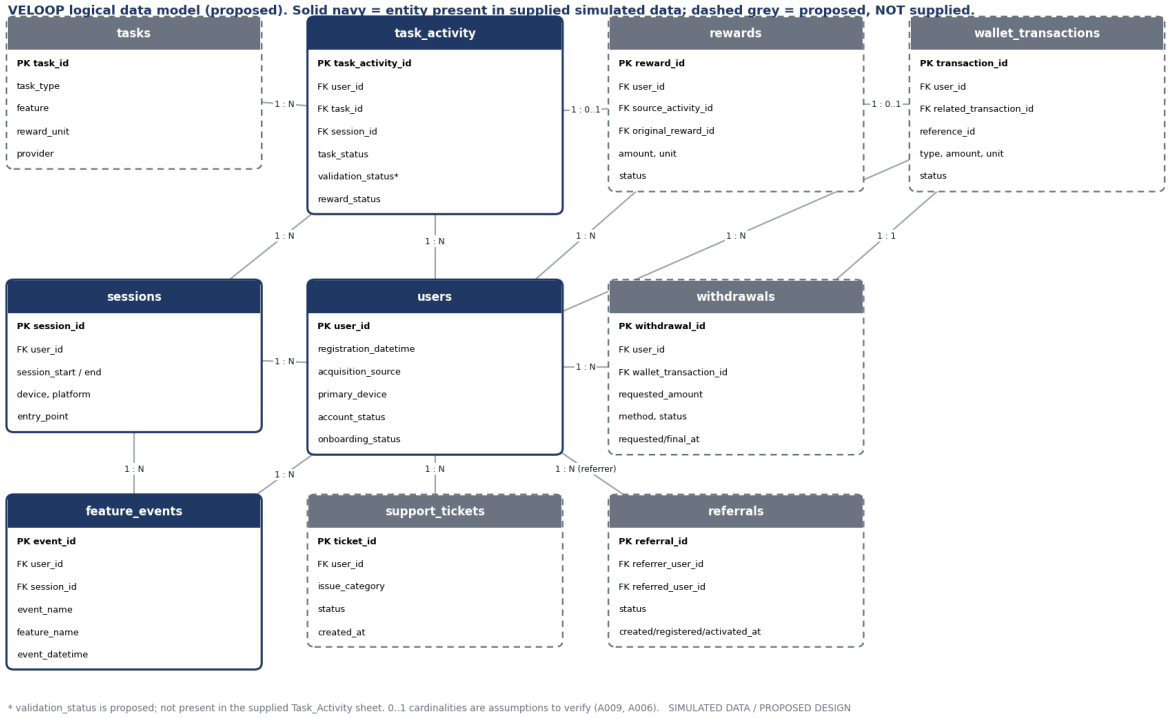


Figure 1. Logical data model (proposed).

Entity	Primary Key	Foreign Keys	Grain
users	user_id	-	One row per account
sessions	session_id	user_id>users	One session per user visit (TBV)
tasks	task_id	-	Catalogue
task_activity	task_activity_id	user_id>users; task_id>tasks	One attempt
rewards	reward_id	user_id>users; source_activity_id>task_activity (or other); original_reward_id>rewards; transaction_id>wallet_transactions	One reward record
wallet_transactions	transaction_id	user_id>users; related_transaction_id>wallet_transactions	Ledger entry
withdrawals	withdrawal_id	user_id>users; wallet_transaction_id>wallet_transactions	Payout request
referrals	referral_id	referrer_user_id>users; referred_user_id>users	Referral
feature_events	event_id	user_id>users; session_id>sessions	Event stream
support_tickets	ticket_id	user_id>users	Ticket
users 1:N task_activity	A user has many attempts; each attempt has one user		
tasks 1:N task_activity	A task catalogue item has many attempts		
task_activity 1:0..1 rewards	An attempt yields at most one reward (assumption; TBV)		
rewards 1:N rewards (self)	A reward can have reversal/correction rows via original_reward_id		
rewards 1:0..1 wallet_transactions	A credited reward has one ledger credit		
wallet_transactions 1:N wallet_transactions (self)	Reversals link to original		
users 1:N withdrawals	-		
withdrawals 1:1 wallet_transactions	Debit entry (assumption; TBV)		
users M:N users via referrals	Referrer/referred; each referred user at most one referrer		
users 1:N feature_events	-		
sessions 1:N feature_events	-		
users 1:N support_tickets	-		

Relationships: users 1:N task_activity, sessions, feature_events, rewards, wallet_transactions, withdrawals, support_tickets; tasks 1:N task_activity; sessions 1:N feature_events; task_activity 1:0..1 rewards; rewards 1:0..1 wallet_transactions; withdrawals 1:1 wallet_transactions (debit entry); rewards and wallet_transactions each relate to themselves for reversal chains; referrals link two users (referrer, referred), many-to-many through users. The 0..1 cardinalities are assumptions to verify.

10. KPI catalogue

46 KPIs across user, engagement, reward, wallet, withdrawal, referral, support/content and data-quality areas. No targets are set; each KPI has a reproducible definition, formula, required fields, data source, frequency, stakeholder, purpose, interpretation, limitation and status. Example: Daily Active Users = unique users generating a qualifying activity event within a UTC calendar day.

KPI ID	KPI Name	Area	Definition	Frequency
K01	Registrations	User	Count of accounts with a registration timestamp in the period	Daily
K02	Registrations by source	User	Registrations grouped by registration_source	Weekly
K03	Daily Active Users (DAU)	User	Unique users generating a qualifying activity event within a calendar day (UTC)	Daily
K04	Returning users	User	Active users in the period who also had qualifying activity in an earlier period (or registered earlier)	Weekly
K05	Weekly/Monthly Active Users	User	Unique users with qualifying activity in 7/30 days	Weekly/ Monthly
K06	Onboarding completion rate	User	Registered users who completed onboarding / registered users	Weekly
K07	Inactive users	User	Registered, non-closed users with no qualifying activity for N days (N defined by Product)	Weekly
K08	Stickiness	User	DAU / MAU for the same month	Monthly
K09	Time to first activity	User	Median time from registration to first qualifying event	Weekly

KPI ID	KPI Name	Area	Definition	Frequency
K10	Account status distribution	User	Users per account_status	Weekly
K11	Task starts	Engagement	Task attempts with a start timestamp	Daily
K12	Task completion rate	Engagement	Validated completions / starts	Daily
K13	Task completions	Engagement	Count of validated completions	Daily
K14	Task failure rate by category	Engagement	Failed attempts by failure_category / starts	Weekly
K15	Feature usage	Engagement	Distinct users using each feature	Weekly
K16	Repeat usage rate	Engagement	Users with >=2 activity days in a feature / users who used it	Weekly
K17	Feature adoption rate	Engagement	Users who used a feature at least once / active users	Monthly
K18	Reward transactions	Reward	Count of reward records created	Daily
K19	Reward credited amount	Reward	Sum of credited reward amounts, per unit	Daily
K20	Reward reversals	Reward	Count and sum of reversed rewards	Weekly
K21	Reward reversal rate	Reward	Reversed / credited rewards	Weekly
K22	Pending rewards	Reward	Rewards in pending status and age	Daily
K23	Reward by feature	Reward	Credited amount per feature/source	Weekly
K24	Activity-to-credit rate	Reward	Validated completions with a credited reward / validated completions	Weekly
K25	Wallet transactions	Wallet	Count of ledger entries	Daily
K26	Wallet credit volume	Wallet	Sum of credit entries completed per unit	Daily
K27	Redemption activity	Wallet	Count/sum of redemption/exchange entries	Weekly
K28	Wallet reversals	Wallet	Reversal entries and share of all transactions	Weekly
K29	Ledger reconciliation gap	Wallet	Credited rewards without matching wallet credit (and reverse)	Daily
K30	Withdrawal requests	Withdrawal	Requests created in period	Daily
K31	Completed withdrawals	Withdrawal	Requests with status completed	Daily
K32	Pending withdrawals (queue)	Withdrawal	Requests in requested/processing at snapshot	Hourly/Daily
K33	Rejected withdrawals	Withdrawal	Requests rejected, by reason_category	Weekly
K34	Reversed withdrawals	Withdrawal	Requests reversed	Weekly
K35	Withdrawal processing time	Withdrawal	Median/P90 of finalized_at - requested_at for completed	Daily
K36	Withdrawal completion rate	Withdrawal	Completed / (completed+rejected+reversed) for requests older than a stated window	Weekly
K37	Withdrawal volume	Withdrawal	Sum of completed final_amount per unit	Daily
K38	Method distribution	Withdrawal	Share of requests per method	Weekly
K39	Referrals created	Referral	Referral records created	Daily
K40	Referral registrations	Referral	Referrals with registered_at	Daily
K41	Referral activations	Referral	Referrals with activated_at	Daily
K42	Referral conversion rate	Referral	Converted / created referrals (cohorted)	Weekly
K43	Support requests	Support/Content	Tickets created	Daily
K44	Issue categories	Support/Content	Tickets by issue_category	Weekly
K45	Help-center engagement	Support/Content	Help article views and users per category	Weekly
K46	Data quality pass rate	Data Quality	Records passing all hard rules / records loaded	Daily

11. Simulated data profiling (Work Package J)

All statements describe the supplied simulated datasets only. Reference period stated in the files: 1 April to 15 September 2026.

dataset	records	columns	earliest	latest	duplicate_rows	missing_cells
A:Users_1000_Clean	1000	27	2026-04-01 02:10:00	2026-09-17 10:08:00	0	953
A:Users_Messy_Audit	1008	28	2026-04-01 02:10:00	2026-09-17 10:08:00	8	966
B:01_Users_1000	1000	16	2026-04-01 02:00:41	2026-09-15 20:26:50	0	846
B:02_Sessions	7913	12	2026-04-02 20:14:28	2026-09-15 22:14:12	0	466
B:03_Feature_Events	35704	13	2026-04-02 20:14:31	2026-09-15 22:13:50	0	0
B:04_Daily_Activity	6667	17	2026-04-02 00:00:00	2026-09-15 00:00:00	0	0
B:05_Task_Activity	9846	12	2026-04-02 20:14:41	2026-09-15 21:59:48	0	10494
B:06_Milestones	4799	6	2026-04-02 08:18:00	2026-09-15 22:13:28	0	40
B:10_Event_Data_Quality_Audit	35714	13	2026-04-02 20:14:31	2026-09-15 22:13:50	10	0

Relationship tests

parent	child	child_rows	orphans	parents_without_children	result
B:01_Users_1000	B:02_Sessions	7913	0	16	PASS
B:01_Users_1000	B:03_Feature_Events	35704	0	16	PASS
B:01_Users_1000	B:04_Daily_Activity	6667	0	16	PASS
B:01_Users_1000	B:05_Task_Activity	9846	0	16	PASS
B:01_Users_1000	B:06_Milestones	4799	0	0	PASS
B:01_Users_1000	B:10_Event_Data_Quality_Audit	35714	0	16	PASS
B:02_Sessions	B:03_Feature_Events	35704	0	302	PASS

parent	child	child_rows	orphans	parents_without_children	result
B:02_Sessions	B:05_Task_Activity	9846	0	3301	PASS
B:02_Sessions	B:06_Milestones	4799	0	4665	PASS
A:Users_1000_Clean	B:02_Sessions	7913	0	16	PASS
A:Users_1000_Clean	A:Users_Messy_Audit	1008	0	0	PASS

Result: every user_id and session_id reference resolves (0 orphans). 16 users have no sessions, events or tasks. The cross-file test passes on IDs but fails on content: the two Users tables disagree (see below).

Cross-file conflict between the two Users tables

field	users_where_A_differs_from_B	of
full_name	1000	1000
email	1000	1000
age	960	1000
gender	655	1000
state	945	1000
city	977	1000
registration_datetime	1000	1000
acquisition_source	829	1000
account_status	571	1000
onboarding_status	463	1000
primary_device	492	1000
referral_code	1000	1000

KPI test: can the proposed KPIs be calculated from what was supplied?

kpi	name	calculable	limitation_or_data_required
K01-K02	Registrations (by source)	YES	Needs one master users table
K03-K05	DAU / WAU / MAU	YES (proxy)	Qualifying-event list still to be defined
K04	Returning users	YES	
K06	Onboarding completion rate	PARTIAL	Snapshot only; no onboarding events or timestamps
K07	Inactive users	YES	As-of date undocumented
K09	Time to first activity	YES	
K10	Account status distribution	YES	No status history
K11-K13	Task starts / completions / completion rate	YES	No validation_status; 648 Pending
K14	Failure rate by category	PARTIAL	Reasons contradict status in some rows
K15-K17	Feature usage / adoption / repeat	YES	
K18-K20,K23	Reward transactions / amounts / by feature	NO	Only reward_status; counts of credited tasks possible, amounts not
K21-K22	Reversal rate / pending rewards	PARTIAL	No reversal state
K24	Activity-to-credit rate	YES (count only)	Credited equals Completed in every row
K25-K29	Wallet KPIs and reconciliation	NO	No ledger table; only wallet_viewed events
K30-K38	Withdrawal KPIs	NO	Not supplied
K39-K42	Referral funnel	NO	Only referral_code and referral events; no referrer/referred link
K43-K45	Support KPIs	NO	Not supplied
K46	Data-quality pass rate	YES	

12. Data quality audit (Work Package K)

34 issues were found by automated checks: 3 Critical, 11 High, 15 Medium, 5 Low. Each has detection logic, likely cause, correction, prevention and owner in the workbook; the critical, high and most informative issues are listed here.

issue_id	dataset	field	issue_type	records	severity
DQ-001	A:Users_Messy_Audit	user_id	duplicate IDs / duplicate rows	8	High
DQ-005	A:Users_Messy_Audit	registration_datetime	unexpected null in required field	1	Critical
DQ-008	A:Users_Messy_Audit	email_verification_status	unexpected null in required field	1	High
DQ-009	A:Users_Messy_Audit	onboarding_status	unexpected null in required field	1	High
DQ-011	A:Users_Messy_Audit	registration_datetime	malformed date	1	Critical

issue_id	dataset	field	issue_type	records	severity
DQ-012	A:Users_Messy_Audit	registration_datetime	value conflicts with clean baseline	2	High
DQ-015	A:Users_1000_Clean	(table)	personal data present beyond need	1000	High
DQ-016	B:01_Users_1000	referral_code	duplicate IDs (codes should be unique)	2	High
DQ-017	A vs B	user_id (cross-file)	conflicting attributes for identical user_ids	1000	Critical
DQ-018	B:02_Sessions	events_in_session	aggregate cannot be reconciled with Feature_Events	7537	High
DQ-023	B:04_Daily_Activity	completed_task_count	impossible value (completed > total)	423	High
DQ-025	B:05_Task_Activity	task_id	one task_id mapped to several categories	80	High
DQ-027	B:05_Task_Activity	reward amount / unit	missing required analytical field	9846	High
DQ-030	B:10_Event_Data_Quality_Audit	event_id	duplicate events	10	High

Cleaning demonstration

dataset	input_rows	duplicates_removed	quarantined	clean_rows	status_normalised
Users_Messy_Audit	1008	8	9.0	991	-
Event_Data_Quality_Audit	35714	10	-	35704	4.0

Rules: exact duplicates are removed by ID (first kept); rows with a malformed date or a missing required value are quarantined, never imputed; status and feature names are trimmed and normalised to their most frequent spelling. Rows are traceable: input rows equal clean plus quarantined plus removed duplicates.

Data quality control framework (Work Package L)

Rule ID	Rule	Layer	Action on Failure
Q01	Primary key present and unique	Application + Ingestion	Reject at ingestion; quarantine
Q02	ID format	Ingestion	Trim/reject; log
Q03	Allowed status values	Application + Raw validation	Quarantine unknown value; alert if >0
Q04	Timestamp validity	Ingestion	Quarantine; never silently coerce
Q05	Timestamp ordering	Transformation	Flag; exclude from duration KPIs
Q06	Amount validity	Application + Ingestion	Reject
Q07	Unit validity	Transformation/Analytics	Block KPI if mixed
Q08	Referential integrity	Transformation	Orphan table; alert
Q09	Duplicate transaction/event	Ingestion + Raw	Keep first, log duplicates
Q10	Valid state transitions	Application + Transformation	Flag impossible transitions
Q11	Reward-wallet reconciliation	Analytics	Daily reconciliation report (K29)
Q12	Reversal linkage	Transformation	Flag
Q13	Conditional required fields	Application + Transformation	Flag
Q14	Consistent naming/case	Collection + Raw	Normalise in staging; report drift
Q15	Referral stage order	Transformation	Flag
Q16	Freshness	Pipeline monitoring	Alert owner
Q17	Volume anomaly	Pipeline monitoring	Alert (no fixed thresholds invented here)
Q18	Schema drift	Raw	Alert; block load if breaking
Q19	PII/secrets screening	Collection + Raw	Quarantine and escalate
Q20	Dashboard certification	Dashboard	Block uncertified

13. Dashboard blueprint

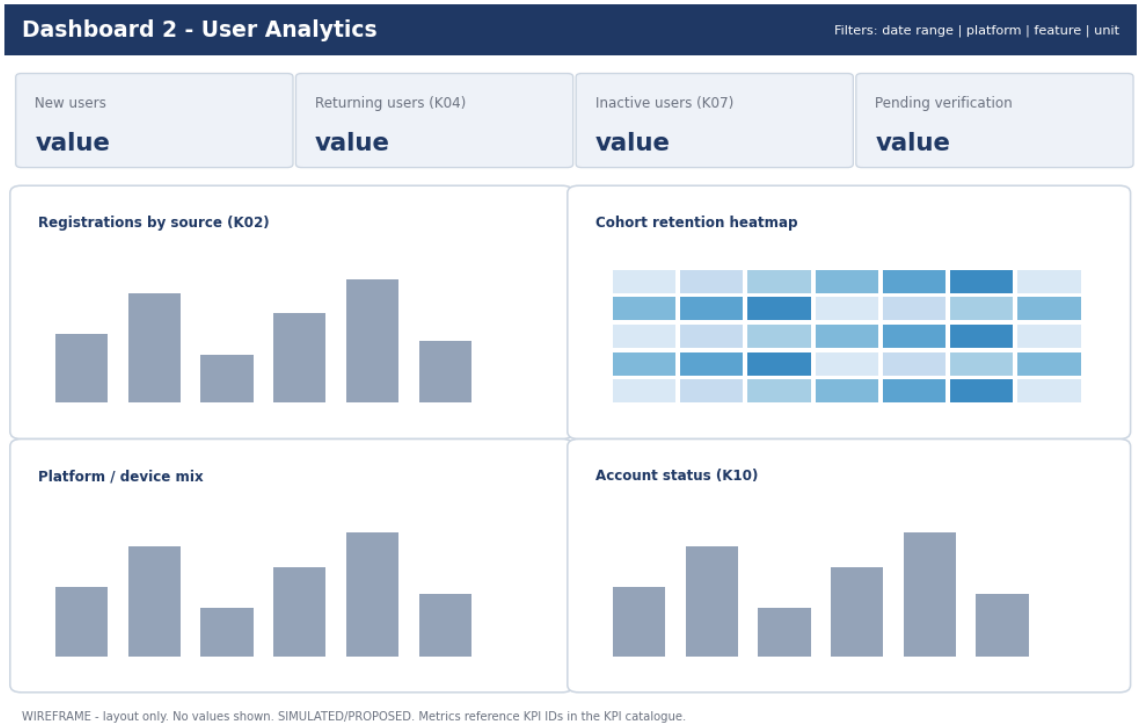
Seven dashboards with 28 visuals, each specified with metric, calculation, source, visualization, filters, dimensions, date range, refresh, business question and decision supported (workbook sheet 11). Wireframes show layout only. They carry no values.

Dashboard	Metric	Visualization	Refresh	Decision Supported
1 Executive Overview	Registrations (K01)	Line + KPI card	Daily	Acquisition focus
1 Executive Overview	DAU/WAU/MAU (K03,K05)	Line + cards	Daily	Product priorities
1 Executive Overview	Task completion rate (K12)	KPI card + trend	Daily	Fix vs expand tasks
1 Executive Overview	Reward credited by unit (K19)	Stacked bar per unit (never mixed)	Daily	Reward economy review
1 Executive Overview	Withdrawal queue + processing time (K32,K35)	Card + trend	Hourly/ Daily	Ops staffing

Dashboard	Metric	Visualization	Refresh	Decision Supported
1 Executive Overview	Referral funnel (K39-K42)	Funnel	Daily	Referral program design
1 Executive Overview	Data quality pass rate (K46)	Card	Daily	Hold/publish decision
2 User Analytics	Registrations by source (K02)	Stacked bar	Daily	Channel focus
2 User Analytics	Returning vs new (K04)	Stacked area	Daily	Retention work
2 User Analytics	Cohort retention	Heatmap	Weekly	Onboarding/feature changes
2 User Analytics	Device/platform and country mix	Bar/table (no map unless country verified)	Weekly	Platform priorities
2 User Analytics	Account status (K10), inactive (K07)	Donut + trend	Weekly	Re-engagement
3 Engagement	Feature usage (K15, K17)	Bar	Daily	Feature roadmap
3 Engagement	Task funnel viewed>opened>started>completed	Funnel	Daily	UX fixes
3 Engagement	Failure categories (K14)	Bar	Daily	Provider/engineering action
3 Engagement	Repeat usage (K16)	Line	Weekly	Habit features
4 Rewards	Credited by source (K23)	Bar	Daily	Reward design
4 Rewards	Reward status mix (K18-K22)	Stacked bar	Daily	Integrity review
4 Rewards	Activity-to-credit (K24)	Line	Daily	Bug triage
5 Wallet	Transactions by type (K25)	Stacked bar	Daily	Monitoring
5 Wallet	Credits vs redemptions (K26,K27)	Line (per unit)	Daily	Economy balance
5 Wallet	Reconciliation gap (K29)	Table (should be empty)	Daily	Incident
6 Withdrawals	Status distribution (K30-K34)	Stacked bar	Daily	Ops process
6 Withdrawals	Processing time (K35)	Line + histogram	Daily	SLA discussion
6 Withdrawals	Method distribution (K38), volume (K37)	Bar	Weekly	Method support
6 Withdrawals	Rejection reasons (K33)	Bar	Weekly	Guidance to users
7 Referrals	Referral funnel (K39-K42)	Funnel	Daily	Program design
7 Referrals	Time created>registered>activated	Box plot	Weekly	Activation rule review



Wireframe: 1 Executive Overview



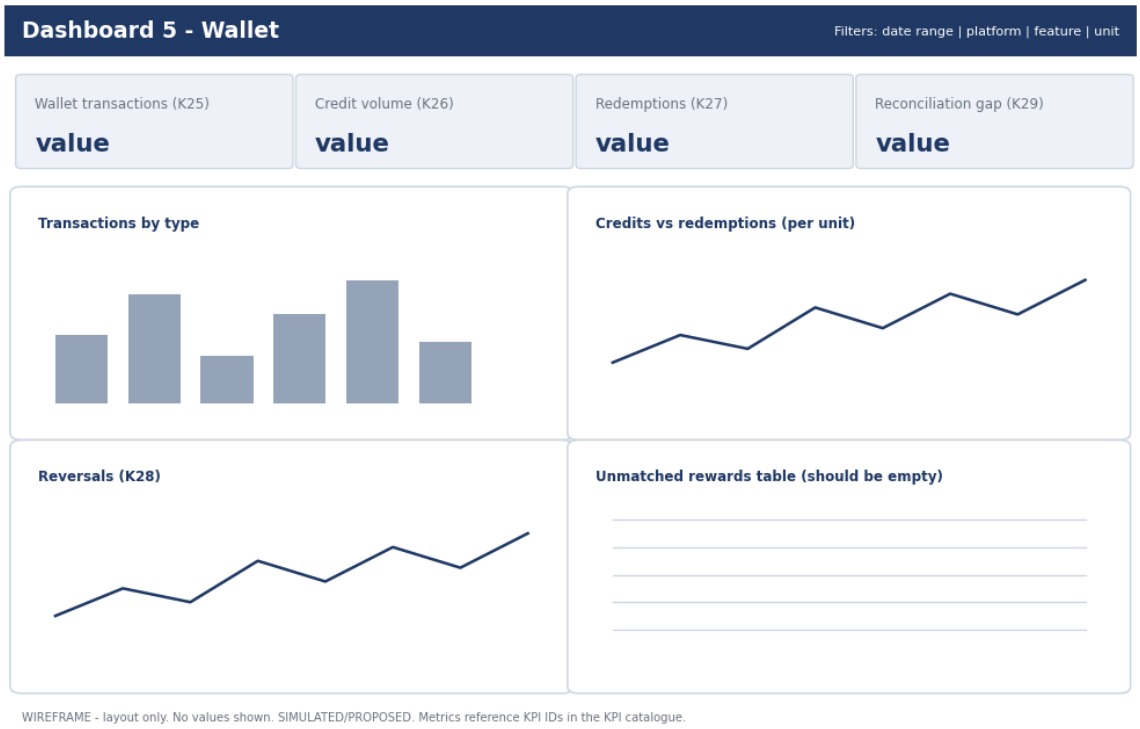
Wireframe: 2 User Analytics



Wireframe: 3 Engagement



Wireframe: 4 Rewards



Wireframe: 5 Wallet



Wireframe: 6 Withdrawals



Wireframe: 7 Referrals

14. Developer analytics handoff

15 requirements (REQ-001 to REQ-015) specify, per event: trigger, payload with type, required flag and validation, deduplication key, timestamp rule, source, why needed, KPI dependency, acceptance test and priority. Example:

Item	Specification
Requirement ID	REQ-004
Event	task_completed
Trigger	Completion has passed applicable validation
Payload	task_activity_id:Y; user_id:Y; task_id:Y; task_type:Y; status:enum(completed):Y; validation_status:enum:Y; reward_amount:decimal:N; reward_unit:enum:N; provider_ref:string:N
Deduplication	event_id; task_activity_id+status
Timestamp	completion time UTC
Acceptance test	When a valid completion occurs, exactly one analytically usable completion event is recorded for that task_activity_id, subject to final engineering idempotency design
Priority	Critical

Requirement ID	Event	KPI Dependency	Priority
REQ-001	user_registered	K01,K02	Critical
REQ-002	user_logged_in	K03,K04	Critical
REQ-003	task_started	K11,K12	Critical
REQ-004	task_completed	K12,K13,K24	Critical
REQ-005	task_failed	K14	High
REQ-006	reward_credited	K19,K23,K24	Critical
REQ-007	reward_reversed	K20,K21	Critical
REQ-008	wallet_transaction_created	K25-K29	Critical
REQ-009	wallet_transaction_reversed	K28	Critical
REQ-010	withdrawal_requested	K30	Critical
REQ-011	withdrawal_processing	K32,K35	Critical
REQ-012	withdrawal_completed	K31,K35,K37	Critical
REQ-013	withdrawal_rejected	K33	Critical
REQ-014	referral_registered / referral_activated / referral_converted	K40-K42	High
REQ-015	onboarding_completed	K06	High

15. Data pipeline and operating model

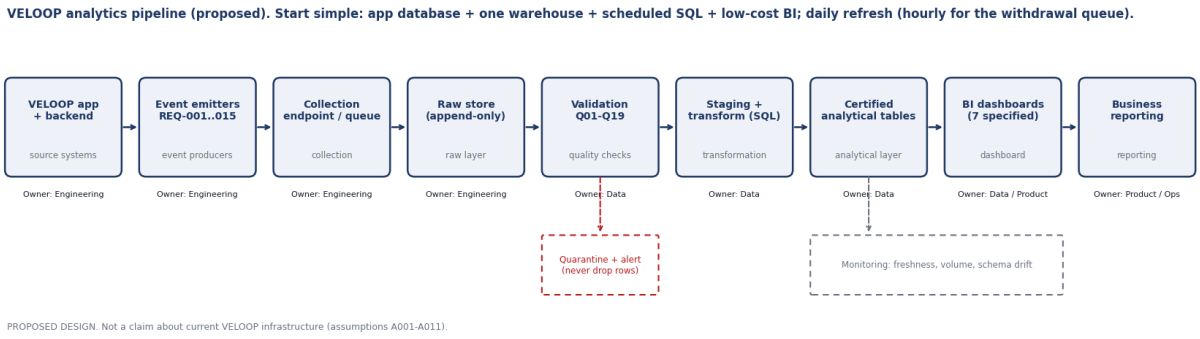


Figure 2. Proposed analytics pipeline with owners, quarantine path and monitoring.

A practical flow, deliberately without expensive infrastructure: VELOOP application and backend > event emitters (REQ-001 to 015) > collection endpoint or queue > raw append-only store > validation (rules Q01 to Q19; failures go to quarantine and alert) > staging and transformation (SQL models) > certified analytical tables > BI dashboards > business reporting, with monitoring on freshness, volume and schema drift.

- **Start small:** the application database plus one managed warehouse (or a relational database) with scheduled SQL transformations and a low-cost BI tool. Move to streaming or heavier tooling only when volume or latency requires it.
- **Refresh:** daily batch; hourly only for the withdrawal queue.
- **Failure handling:** failed loads retried; bad rows quarantined, never dropped; owners alerted.
- **Ownership:** Engineering owns emitters and the raw layer; Data owns validation, transformation, KPI definitions and dashboards; Product and Operations own definitions of qualifying activity and states.

16. Privacy, access and governance

Role	Access Scope	Level	Notes
Engineering	Event emitters, raw/staging, schemas	Full on raw; no credentials in analytics stores	Secrets, keys never enter analytics
Data Analyst	Curated analytical tables, DQ logs	Read curated; read raw only if justified	No restricted fields by default
Product/Operations	Aggregated dashboards	Aggregates; row-level only for ops queues (IDs, no personal data)	provider_ref restricted
Content Team	Aggregated help/content metrics	Aggregates only	-
Finance	Ledger and withdrawal aggregates, reconciliation	Curated ledger views	-
Intern/Trainee	Sanitised/simulated datasets	Approved sanitised datasets only	No raw production
Never collected	Passwords, authentication secrets, API keys, payment credentials, card/bank numbers, private credentials, security-control details, fraud thresholds	-	Block at ingestion (Q19)

Never collected or stored in analytics: passwords, authentication secrets, API keys, payment credentials, card or bank numbers, security-control details, fraud thresholds. Payout provider references are Restricted. The supplied Users data breaks the minimisation rule (DQ-015) and should be reduced to user_id plus coarse attributes in the analytics layer.

Change management

Every change to an event, field, reward mechanic, withdrawal method, KPI or dashboard is logged (date, change, old and new definition, reason, affected KPIs, dashboards and reports, owner, version). Event schemas are versioned; renamed events keep an alias for a declared period; a changed KPI becomes a new version annotated on its dashboards; every new feature passes an analytics checklist (events, dictionary entries, KPIs, quality rules) before launch.

17. Analytics readiness assessment

Rubric (not a company metric): 0 = nothing verified, 1 = designed only, 2 = designed and tested on simulated data, 3 = verified in production. No dimension can reach 3 until production data exists.

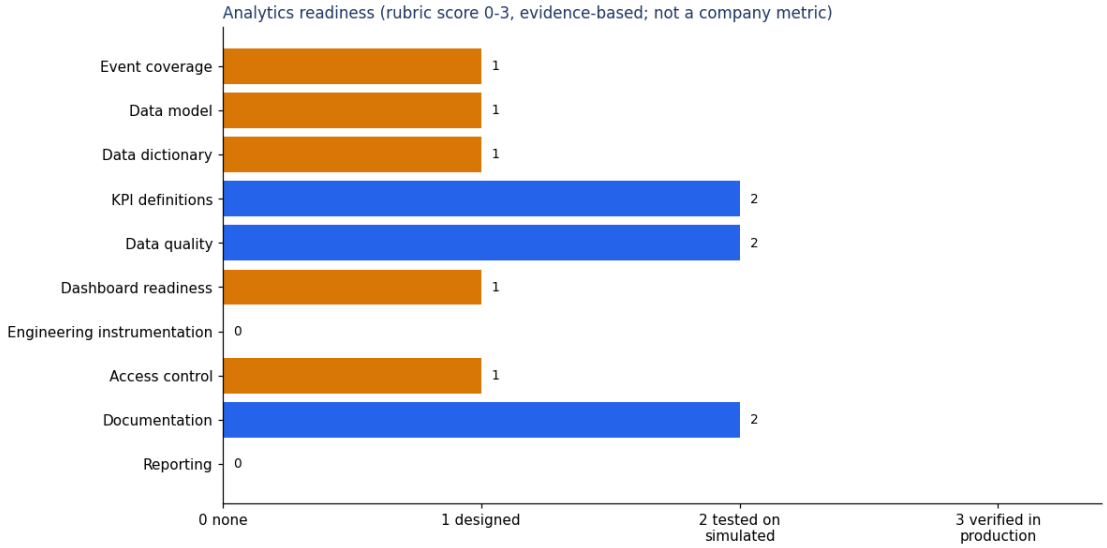


Figure 3. Readiness scorecard.

Dimension	Score	Priority	Evidence
Event coverage	1	Critical	38 events designed; none verified as implemented; supplied events do not follow the catalogue
Data model	1	Critical	Model designed; keys/links tested only for users, sessions, events, tasks. Rewards, wallet, withdrawals, referrals not supplied
Data dictionary	1	High	100 fields proposed; supplied dictionaries cover 2 + 26 fields and miss columns (DQ-033, DQ-034)
KPI definitions	2	High	46 KPIs defined; 18 KPI groups tested against supplied structure (10 calculable, 3 partial, 5 not)
Data quality	2	High	20 rules designed; 34 issues detected by automated checks with 11 passing tests; cleaning rules implemented
Dashboard readiness	1	Medium	7 dashboards specified and wireframed; no certified tables to power them
Engineering instrumentation	0	Critical	Nothing verified: IDs, idempotency, UTC timestamps and event emitters are all unknown (A001-A011)
Access control	1	High	Role model proposed; supplied data contains unnecessary personal fields (DQ-015)
Documentation	2	Medium	README, report, workbook, ER diagram, handoff spec; versioned in this project
Reporting	0	Medium	No governed recurring report exists; depends on production data

Dimension	Priority	Reason
Event coverage	Critical	Without events no behaviour KPI can be produced
Data model	Critical	Joins depend on stable keys
Data dictionary	High	Misread fields cause wrong KPIs
KPI definitions	High	Definitions must be reproducible
Data quality	High	Bad data silently skews reports
Dashboard readiness	Medium	Needs data first
Engineering instrumentation	Critical	No stable transaction_id means no reliable wallet dedup
Access control	High	Restricted fields must not leak
Documentation	Medium	Reuse by next analyst
Reporting	Medium	Depends on above

18. Assumption and verification register

ID	Unknown	Why It Matters	Risk	Verification Owner	Status
A001	Does backend currently emit task completion events?	Required for task analytics	High	Engineering	To Be Verified
A002	Is there a stable user_id independent of email/phone?	Join key	High	Engineering	To Be Verified
A003	Does a session identifier exist?	Session analytics	High	Engineering	To Be Verified

ID	Unknown	Why It Matters	Risk	Verification Owner	Status
A004	Is there a unique transaction_id per wallet entry?	Dedup, reconciliation	High	Engineering	To Be Verified
A005	Which units are withdrawable and how are units converted?	Unit-safe KPIs	High	Product	To Be Verified
A006	What withdrawal states and timestamps are stored?	Lifecycle KPIs	High	Engineering/Ops	To Be Verified
A007	Which withdrawal methods exist?	Method distribution	Medium	Ops	To Be Verified
A008	How is reward validation done for third-party tasks?	Completion trust	High	Engineering	To Be Verified
A009	Are rewards pending before credit?	Pending KPI	Medium	Product	To Be Verified
A010	How is a referral activated/converted?	Referral KPIs	High	Product	To Be Verified
A011	Are timestamps stored in UTC?	Daily KPIs	High	Engineering	To Be Verified
A012	Are Spin, Mining, Tap&Earn, Missions, Badges, Levels live?	Event scope	Medium	Product	To Be Verified
A013	What account statuses exist?	Lifecycle	Medium	Engineering	To Be Verified
A014	Is acquisition source captured?	Acquisition	Medium	Engineering/ Marketing	To Be Verified
A015	What support tool is used and does it export categories?	Support KPIs	Low	Support	To Be Verified
A016	Which Users table is the master (A or B share user_ids but conflict)?	Cross-file integrity	High	Product/Data	To Be Verified
A017	Which events count as 'qualifying activity' for DAU?	DAU definition	High	Product	To Be Verified
A018	Is a legal/privacy review of country and device fields needed?	Governance	Medium	Legal/Product	To Be Verified

19. Implementation backlog

Priority	Requirement	Reason	Dependency	Owner	Stage
Critical	Stable user_id	Join key for all analysis	Auth	Engineering	Prelaunch
Critical	Immutable transaction_id + idempotency on wallet writes	Ledger dedup	Wallet	Engineering	Prelaunch
Critical	Define units and conversion rules	Unit-safe totals	Product decision	Product	Prelaunch
Critical	Withdrawal state + timestamp history	Lifecycle KPIs	Withdrawal service	Engineering	Prelaunch
Critical	Separate reward record from activity record	Activity vs credited	Rewards service	Engineering	Prelaunch
Critical	Emit events REQ-001..013	Behaviour KPIs	Event SDK/backend	Engineering	Prelaunch
High	UTC timestamp standard	Daily KPIs	Platform	Engineering	Prelaunch
High	Ingestion validation rules Q01-Q06,Q09,Q19	Prevent bad data	Pipeline	Data/ Engineering	Prelaunch
High	Task failure classification	Failure analysis	Task service	Product/ Engineering	Prelaunch
High	Referral stage rules	Referral KPIs	Referral service	Product	Prelaunch
High	Approved status/enum lists	Validation	Dictionary	Data/ Engineering	Prelaunch
High	Role-based access to curated tables	Governance	Data platform	Data/ Engineering	Prelaunch
High	Reward-wallet reconciliation job (K29)	Ledger trust	Analytics layer	Data	Prelaunch
Medium	Session identifier	Session analytics	Client SDK	Engineering	Prelaunch/ Postlaunch
Medium	Acquisition source capture	Acquisition	Client/backend	Engineering	Postlaunch
Medium	Support ticket categories export	Support KPIs	Support tool	Support	Postlaunch
Medium	Engagement events for Spin, Missions, Streaks	Feature KPIs	Feature teams	Engineering	Postlaunch
Medium	Dashboards 1,5,6 on certified tables	Management reporting	Production data	Analytics	Postlaunch
Low	Dashboards 2,3,7 refinements, cohort views	Insight	Production data	Analytics	Postlaunch
Low	Badge/level analytics	Progression	Feature live	Analytics	Postlaunch

20. Limitations and decisions needed

- Rewards, Wallet, Withdrawals and Referrals datasets were not supplied: related KPIs and tables are design-only.
- The behaviour workbook labels itself 'Task 02'. It was used as additional simulated data because it overlaps the Task 01 scope; please confirm.
- Decision needed: which Users table is the master (A016).
- Nothing is marked Confirmed beyond the brief and the supplied files' own READMEs; engineering verification is required before any proposed event or field is treated as real.
- When production data exists, the simulated datasets must be replaced by governed production data and never silently mixed with it.

Integrity checklist (brief section 34)

Rule	Status
Never fabricate production data	Met
Never represent simulated data as actual VELOOP statistics	Met: every sheet and section is labelled
No fake company performance claims	Met: no targets, no success rates
No invented security mechanisms or fraud thresholds	Met
No credentials requested	Met
No proposed event claimed to exist	Met: all 'Proposed - To Be Verified'
Uncertainty not hidden	Met: 18-item register, Section 20
No unnecessary personal-data fields	Met in design; supplied data violates it (DQ-015)
Conclusions traceable to dataset, source or assumption	Met: issue IDs, KPI IDs, assumption IDs